

# Mahmoud Mohamed El-Saeed

## Junior AI & Machine Learning Engineer

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### Profile Summary

Junior AI / Machine Learning Engineer with hands-on experience developing AI solutions across Computer Vision, NLP, Generative AI, Multimodal AI, and Agentic AI. Skilled in Python, YOLO, RAG, transformer-based models, and AI agents, with experience building and deploying end-to-end ML and AI systems using FastAPI, Hugging Face, and Raspberry Pi. Experienced in developing practical solutions across Machine Learning, Deep Learning, Generative AI, and Agentic AI applications. Passionate about building scalable intelligent systems and continuously developing expertise in AI/ML.

### Experiences

#### AI & Data Science — Agentic AI and Generative AI System Developer 07/2026 – Present

Digital Egypt Pioneers Initiative (DEPI)

- Developed Generative AI applications using LLMs, APIs, and open-source models.
- Built RAG and memory-based systems for context-aware AI applications.
- Implemented Agentic AI and Multi-Agent Systems for task automation and orchestration.
- Worked with Hugging Face models, advanced prompting, and LLM integration.
- Developed and deployed AI applications with monitoring and evaluation.
- Completed a Capstone Project applying Generative AI and Agentic AI concepts.

#### AI / Machine Learning Intern — Uneeq Interns Program 09/2025 – 10/2025

- Built machine learning models using Python and Scikit-learn.
- Performed data preprocessing and exploratory data analysis.
- Applied NLP techniques for text classification tasks.
- Evaluated model performance using standard ML metrics.

#### Artificial Intelligence — National Telecommunication Institute (NTI) 03/2025 – 05/2025

Creativa Mansoura Hub

- Implemented machine learning models using Python and Scikit-learn.
- Performed data preprocessing, feature engineering, and exploratory data analysis.
- Built classification models and evaluated performance using accuracy and ROC-AUC metrics.

### Technical Skills

- **Languages:** Python, SQL, Java, C#, C++, C
- **Deep Learning:** CNN, YOLO (YOLOv8), Transformers, DistilBERT, EfficientNet-B3
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Neural Networks
- **NLP & GenAI:** RAG, FAISS, Ollama, TF-IDF, LoRA Fine-tuning
- **Data Analysis:** Pandas, NumPy, Data Cleaning, Feature Engineering, EDA
- **Data Visualization:** Matplotlib, Seaborn
- **Tools & Deployment:** Git, FastAPI, Streamlit, Docker basics, Hugging Face Spaces
- **Cloud:** AWS (Educate ML Foundations)
- **Embedded / Edge:** Raspberry Pi 5 edge deployment

### Achievements

- **IEEE Xplore Publication & ITC 2026:** Research derived from the graduation project “Intelligent Agricultural Robot for Precision and Sustainable Farming Using AI and IoT” was accepted for publication in IEEE Xplore and presentation at ITC 2026, with project support from ITIDA under its graduation projects support program.
- **Top 3 Student — ML & AI Discovery Camp:** Ranked among the Top 3 Students after completing practical projects in End-to-End ML, NLP, Sentiment Analysis, Customer Ticket Classification, and RAG-based AI systems.
- **3rd Place — ICPC College Contest:** Achieved 3rd place in the college-level ICPC programming contest in the first participation, qualifying for the next stage in Alexandria.

## Projects

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**AI-Powered Plant Disease Detection (Graduation Project)**

Completed — Discussed 27/06/2026 (6 months)

*YOLOv8s | PyTorch | FastAPI | Raspberry Pi 5 | Hugging Face Spaces*

- Built an object detection system covering 44 disease classes across 12 crop species using YOLOv8s.
- Designed and iterated custom dataset-balancing and object-level copy-paste augmentation pipelines in Python (Albumentations) to fix class imbalance.
- Deployed a cloud inference service with FastAPI on Hugging Face Spaces, returning result image URLs.
- Implemented real-time edge inference on a Raspberry Pi 5 for offline/field use.
- Prototyped a companion Flutter mobile UI to display multi-detection results with severity and treatment recommendations.

**AI-Powered Customer Support System**

01/2026 – 02/2026

*NLP | RAG | FAISS | Ollama | Scikit-learn | Streamlit*

- Developed a multi-model AI system to automate customer support ticket processing.
- Implemented a RAG pipeline using FAISS and Ollama LLM to generate contextual responses.
- Built classifiers for ticket category, sentiment, and priority prediction.
- Trained on ~20K support tickets, achieving 97% classification accuracy.
- Built an interactive web interface with dark/light mode using Streamlit.

**Email-Spam-AI-Visual-Security-System**

11/2025 – 01/2026

*DistilBERT | EfficientNet-B3 | SGD | OCR*

- Developed a multimodal spam detection pipeline analyzing both text and images.
- Implemented DistilBERT for text classification, achieving 99% accuracy on the validation dataset.
- Integrated EfficientNet-B3 and Tesseract OCR for image-based spam detection.
- Designed a decision engine combining model outputs to classify emails into Spam, Ham, or Suspicious.

**Customer Churn Prediction Model**

10/2025 – 11/2025

*Random Forest | SMOTE | Scikit-learn*

- Built a machine learning model to predict customer churn.
- Performed data preprocessing, feature engineering, and class balancing using SMOTE.
- Achieved 98% recall and an AUC-ROC score of 0.95.
- Conducted exploratory data analysis to identify key churn drivers.

## Education

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**Bachelor of Computer Engineering — Misr Higher Institute for Engineering and Technology**

09/2021 – 07/2026

*Dakahlia, Mansoura, Egypt*

- **Final year overall grade:** Excellent 94.66%.
- **Cumulative grade:** Very Good with Honors 83.65%.
- **Relevant coursework:** Machine Learning, Artificial Intelligence, Computer Vision, Neural Networks, Data Structures, Algorithms, Database Systems.
- **Graduation project:** AI-powered plant disease detection using YOLOv8s.

## Certifications

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- **Machine Learning Track** — Generative Technology Center (GTC) (Sep 2025)
- **Getting Started with Deep Learning** — NVIDIA (Sep 2025)
- **Machine Learning Internship** — UneeQ Interns (Sep 2025)
- **AWS Educate Machine Learning Foundations** — Amazon Web Services (Aug 2025)
- **HCIA-AI V3.5** — Huawei (May 2025)

## Languages

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**Arabic:** Native / Bilingual

**English:** Proficient